

Operator Discovery Report: candidate_ranker_transfer_crossing_span_balanced

- Operator type: candidate_ranker.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Deterministically re-ranks local-search moves by emphasizing 2-opt crossings and span reduction, while lightly penalizing nearest-neighbor trap moves to improve transfer across clustered, grid-like, and bottleneck instances.
- Novelty classification: delta_weighted_ranker.
- Rediscovery signature: candidate_ranker:delta=1.000,cross=1.000,span=0.400,trap=0.200.
- Surviving candidate: False.
- Problem class where it helps: none confirmed.

Pseudocode

- For each candidate move, compute a composite score from immediate delta, crossing removal, span, trap awareness, and local structure.
- Sort candidate moves by ascending composite score.
- Apply the first move that yields a genuine tour improvement.

Complexity

- Non-empty lines: 13.
- Complexity score: 0.46.
- Fingerprint: 86056cf953d289b7.

Why It Should Help

- The operator changes which local-search moves are tried first, so it can improve search efficiency without rewriting the underlying solver.

Failure Cases

- Underperforms on grid_like_tsp with gap delta 0.054436.
- Does not transplant cleanly into random_restart_2opt.
- Does not transplant cleanly into clustered_local_search.
- No validation family shows a clear improvement over the host baseline.
- Disabling the operator does not hurt, so the ablation does not support a real operator effect.

Transplant Results

- nearest_neighbor_2opt gap delta 0.001527, runtime delta 88.226114, distance-eval delta -100.571429.
- cheapest_insertion_2opt gap delta 0.0, runtime delta 688.245086, distance-eval delta 0.0.
- random_restart_2opt gap delta 0.006029, runtime delta -45.924557, distance-eval delta -2292.571429.
- sparse_three_opt gap delta 0.0, runtime delta 4.833871, distance-eval delta 0.0.
- clustered_local_search gap delta 0.007422, runtime delta 13.279543, distance-eval delta 1542.285714.

Instance-Family Results

- heldout_tsplib operator gap 0.236585 vs baseline 0.235058.
- clustered_tsp operator gap 0.110853 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.003204 vs baseline 0.003204.
- grid_like_tsp operator gap 0.161132 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.0 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.109057 vs baseline 0.109057.
- uniform_euclidean operator gap 0.0 vs baseline 0.0.

Pareto Results

- Gap delta: 0.006293.
- Runtime inflation: 0.019176.
- Distance-evaluation inflation: 0.003718.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator -0.006293.
- simplified_delta_ranker mean gap 0.127067 and delta vs full operator -0.000543.
- shuffled_ranker mean gap 0.121771 and delta vs full operator -0.005839.